

CSA 0378-DATA STRUCTURES

19. HEAP SORT

```
main.c
1 #include <stdio.h>
2
3 void heapify(int a[], int n, int i) {
4     int largest = i, l = 2*i+1, r = 2*i+2, temp;
5
6     if(l < n && a[l] > a[largest])
7         largest = l;
8
9     if(r < n && a[r] > a[largest])
10         largest = r;
11
12     if(largest != i) {
13         temp = a[i];
14         a[i] = a[largest];
15         a[largest] = temp;
16
17         heapify(a, n, largest);
18     }
19 }
20
21 int main() {
22     int a[] = {40,20,10,80,60};
23     int n = 5, i, temp;
24
25     for(i = n/2-1; i >= 0; i--)
26         heapify(a,n,i);
27
28     for(i = n-1; i > 0; i--) {
29         temp = a[0];
30         a[0] = a[i];
31         a[i] = temp;
32
33         heapify(a,i,0);
34     }
35
36     printf("Sorted Array: ");
37     for(i = 0; i < n; i++)
38         printf("%d ", a[i]);
39
40     return 0;
41 }
```

Output

Sorted Array: 10 20 40 60 80

=== Code Execution Successful ===